Homework — 1.4.3 Boolean Algebra

Name: ______________________ Date: __________ Mark: ______ / 20

OCR H446 — set after the 1.4.3 lesson. SHOW ALL WORKING.

Part A — This week's topic (~14 marks)

1. Match each Boolean law to its identity: write the correct letter in the **Match** column. One letter is not used. **[2]** *(AO1)*

Law	Match
Double negation	
De Morgan's law	
Distribution	
Association	
Commutation	

Letter	Identity
A	¬(X · Y) = ¬X + ¬Y
B	X · (Y + Z) = X·Y + X·Z
C	X + Y = Y + X
D	(X + Y) + Z = X + (Y + Z)
E	¬(¬X) = X
F	X + ¬X = 1

2. Complete the truth table for the expression **Q = (A + B) · ¬C**. **[3]** *(AO2)*

A	B	C	A+B	¬C	Q
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

3. A student simplifies **Y = ¬(A + ¬B)**. Their working, shown below, contains **two errors**. Identify and correct each error, then write the correct simplified expression for Y. **[3]** *(AO2)*

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Step 1 (De Morgan's law):   Y = ¬A + ¬(¬B)
Step 2 (double negation):   ¬(¬B) = ¬B
Step 3 (result):            Y = ¬A + ¬B
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4. Complete the paragraph using words from the word bank. Each word may be used at most once; **three** words are not needed. **[3]** *(AO1)*

A half adder adds two single bits. Its Sum output is A ___________ B and its Carry output is A ___________ B. A half adder cannot add the middle columns of a multi-bit binary addition because it has no ___________ input, so a ___________ adder — which accepts three inputs — is used for those columns instead. A D-type flip-flop stores ___________ and its output is only able to change on a ___________.

Word bank: XOR · AND · OR · carry-in · overflow · full · one bit · one byte · clock pulse

5. An alarm circuit has inputs **A**, **B** and **C** and sounds when its output **X = 1**. The truth table below shows X.

A	**B**	**C**	**X**
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

(a) Complete the **Karnaugh map** for X. **[1]** *(AO2)*

	BC = 00	**BC = 01**	**BC = 11**	**BC = 10**
A = 0				
A = 1				

(b) Draw the groups on your Karnaugh map and use them to write the **simplified Boolean expression** for X. **[2]** *(AO2)*

Part B — Retrieval: keep it fresh (~6 marks)

6. *(1.3.1)* Explain the difference between **symmetric** and **asymmetric** encryption, and state **one** advantage of asymmetric encryption. **[3]** *(AO1)*

7. *(1.2.1)* Describe how the **round robin** CPU scheduling algorithm works, and state **one** advantage of it. **[3]** *(AO1)*